

BIO4556
Freshwater Ecology
Comparative Analysis of the St. Hippolyte Lakes

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Abstract	3
1. Introduction.....	3
2. Materials and Methods	4
2.1 Data Collection	4
2.1.1 Water Column Data.....	4
2.2 Analysis of samples	5
2.2.1 Chlorophyll-a Analysis.....	5
2.2.2 Colour	5
2.2.3 Alkalinity	6
2.2.4 Colorimetry	6
3. Results.....	6
3.1 Lac Cromwell	6
3.2 Lac Croche	7
3.3 Lac Triton	7
3.4 Lac Geai	8
4. Discussion	9
4.1 Interpretation	9
4.1.1 Lac Cromwell.....	9
4.1.2 Lac Croche	10
4.1.3 Lac Triton.....	11
4.1.3 Lac Geai	12
4.2 Limitations of the Study	13
Appendix	15

Abstract

4 freshwater lakes are observed and analyzed for their different chemical, spectral, and thermal qualities. These lakes, Lac Croche, Cromwell, Triton, and Geai are located around the St. Hippolyte Station de Biologie des Laurentides, Quebec. Lac Croche was identified as an Oligotrophic Lake, however with the analysis conducted, mesotrophic qualities were identified. Lac Triton possessed the characteristics of a mesotrophic lake, and Lac Geai and Cromwell were seemingly eutrophic. All lakes with the exception of Lac Triton were observed to possess a clinograde oxygen-profile. Lac Triton possessed unique qualities unlike the others observed, with vertical profiles that remained constant.

1. Introduction

These lakes consisted of Lac Cromwell, Croche, Triton, and Geai. These lakes are located at the Station de Biologie des Laurentides, where they were in a walkable proximity from the main station. These lakes are located in the Laurentides within southern Quebec. This study was conducted on September 19, 2025 between the end of the summer and beginning of fall. Outside temperatures at the time of sampling was approximately 10 degrees Celsius. The data collected were for chemical, thermal, and spectral analysis to aid in supporting the characteristics of certain lakes such as oligotrophic, mesotrophic, or eutrophic lakes (Blais, 2025).

Lac Cromwell is a dystrophic lake that possess a high amount of dissolved organic matter, with fulvic humic acids. Lac Croche is a softwater lake that is characterized as an Oligotrophic lake, which possess low productivity, receiving its water mainly by rainfall. Lac

Triton is a characteristically mesotrophic lake and the most shallow lake. Lastly, Lac Geai is a small dystrophic and bog lake with an abundance of sphagnum moss surrounding the area.

The objective of this study is to investigate the distinct features of an oligotrophic, mesotrophic, and eutrophic lake. Additionally, the study is done to observe and distinguish key differences between lakes that are in a similar area, with sediments, temperatures, and elevation.

Main nutrients such as phosphorus and nitrogen levels are especially observed due to their importance in limnology. Where total nitrogen and total phosphorus relationships were formed based on their corresponding concentrations and their dependence on one another (Guildford & Hecky, 2000).

2. Materials and Methods

The St. Hippolyte area is located in the Laurentians of Quebec, Canada. The area of interest consisted of 4 lakes with varying conditions.

2.1 Data Collection

The following data collection is done for water column data. Where a Secchi disk, Multi-probe system, Van Dorn bottle, water sample collection, and light meter are used for water column data.

2.1.1 Water Column Data

The Secchi disk is designed to measure the relative depth of light presentation (Blais, 2025) and it is used by submerging the disc into water until its white quadrats are no longer visible, measuring the depth until visibility stops. The second measurement is then taken by raising the disk until the white portion appears again. This is conducted on the shaded side of the boat to ensure clarity and better visibility of the Secchi disk. The measurements that are obtained

are then used to characterize the type of lake. The Van Dorn bottle is utilized in order to collect water samples at different depths of the water column. This is done by a trigger device that closes the bottle at light meter is used to measure the light intensity of a water column at one-meter intervals. The mechanism is designed to send a metal messenger that is dropped down the rope that is attached to the sampler to close the mechanism doors, this mechanism allows for the collection water at the desired depth. A measurement is made per meter of depth at each lake respectively. A light meter is an instrument used to measure light intensity at desired depths. Readings are taken per meter at each lake. The device reads values in micro-mols per meter squared second. This method allows the calculation of the light extinction coefficient for each meter of depth at each lake.

2.2 Analysis of samples

2.2.1 Chlorophyll-a Analysis

The water samples collected via Van Dorn contraption were used to analyze chlorophyll-a concentrations. This was done by filtering 250 mL of the sample water through a fibre glass filter and placing the filter into a 15 mL conical tube where 10 mL of 95% ethanol was then added into the tube. This was done to all samples and was wrapped in aluminum foil and stored in a dark and cool place for approximately 16 hours to extract the chlorophyll from the filter. After the wait time, the samples were centrifuged for ten minutes, placed into a cuvette that was rinsed with 95% ethanol, and analyzed in a fluorometer to measure the concentration of chlorophyll-a.

2.2.2 Colour

Using the sampled filtered lake water from the chlorophyll-a analysis, the colour was analyzed through a spectrophotometer with a wavelength set at 440nm. The spectrophotometer cuvette is filled with distilled water, wiped with a Kimwipe and used for the blank sample to zero for each

reading. Another cuvette is primed and filled by lake water samples and absorbance values were measured.

2.2.3 Alkalinity

In order to measure the alkalinity of lake water samples, acid is added dropwise into the sample to achieve neutrality. This was done by adding 100 mL of lake water into a beaker with a magnetic stir bar, a stir plate is turned on in order to maintain constant mixing of the substrate. A pH probe was inserted into beaker, and 0.1 N HCl was added in 10 μ l increments until the mixture reaches a pH of 3.8. The pH is recorded at the inflection point, and the alkalinity was then calculated for each corresponding water sample

2.2.4 Colorimetry

The HACH machine is used in order to analyze substrate concentrations of total iron, ferrous iron, nitrate, sulfide, and sulfate in the lake waters. These were conducted by adding reagents of into 10 mL samples of the lake water to react with. Lake samples of each depth were sampled and concentrations were measured accordingly based off of each substrate type measured. For every substrate besides nitrate, a blank of the lake water was used in order to calibrate the machine or analysis.

3. Results

3.1 Lac Cromwell

Based on the multifigure for Lac Cromwell, the thermal characteristics represented a decrease in temperature from four meters below surface temperatures to the lake floor. By observing the spectral characteristics, Chlorophyll-a concentration increased from three meters to seven meters below surface level. Additionally, colour absorption had remained consistent around 0.09 abs between surface levels and down to five meters, then increased from five to seven meters to 0.42 abs. Oxygen concentration and saturation had both decreased past four meters below the surface

level. Alkalinity and pH had both remained constant around 6.5-8.5 mg/L and 5.8-6.8 respectively. All substrates measured through the HACH machine were represented to have increase with depth, with the exception of sulfate, which had only detected a concentration on the surface of the lake.

3.2 Lac Croche

Based off of the thermal measurements, water temperature drops at a depth of 4m below the surface by approximately ten degrees Celsius. When observing the spectral characteristics that were measured, chlorophyll-a concentrations appear to be relatively low with the exception of an anomaly data point at seven meters below surface levels. Colour absorption measurements at 440 nm conducted represent values between 0 and 0.1 from the surface level and down to approximately seven meters in lake depth. Past seven meters, the absorption value increases. Oxygen saturation and concentrations both decrease at approximately five meters below surface levels. Conductivity levels had increased at approximately seven meters below the lake surface. The pH values recorded at Lac Croche remained consistent throughout the depth of the lake at a pH of ~6. The Alkalinity procedure that was conducted yielded variable concentrations that gradually increased with depth. The chemical characteristics measured through the HACH machine revealed relatively constant values in depth for ferrous iron, sulfate, and nitrate. Total phosphorus concentrations gradually increased from 3.4 ug/L upwards to 42 ug/L from the surface and down to 11 meters. The sulfide concentrations appeared to decrease with depth from 9.0 to 3.0 ug/L. Lastly, total iron had increased in depth, where values had noticeably increased at 7 meters below surface level.

3.3 Lac Triton

From the multi-figure of Lac Triton, thermal characteristics were represented by a constant temperature throughout its vertical profile. Spectral characteristics are represented by both

chlorophyll-a concentration and colour absorption. These values yielded stable and constant values per its corresponding water depths. Oxygen concentration and saturations were also both remaining constant in the ~7.0 mg/L and 75-79% range. Additionally, conductivity was also constant across the Lac Triton's vertical profile at 16.2 uS/cm. Alkalinity and pH also remained consistent, where the alkalinity conducted yielded consistent values between 4.0-5.0 mg/L and measurements were constant at 6.5-6.7 pH. The total phosphorus and sulfate concentrations that were analyzed in the HACH machine remained relatively stable throughout the lakes profile. Total phosphorus and ferrous iron showed gradual increase in concentration with depth. Total phosphorus represents a small increase, values ranging from 7.6-9.0 ug/L. Ferrous iron had an increase in depth from 0.01 ug/L detected at the surface to 0.06 ug/L at mid sections of the lake, and no detection at the bottom of the lake. Total iron, nitrate, and sulfate all show consistent values. Total iron remained between 0.03 and 0.06 ug/L, nitrate concentrations stayed within 0.30 and 0.34 ug/L, there was no detection of sulfate concentration at all depths.

3.4 Lac Geai

The temperature profile of Lac Geai is represented by a decrease in depth. Where temperatures noticeably drop at 4 meters. The surface temperature observed was 15.9 degrees Celsius, dropping down to 5.3 degrees at the bottom of the lake. Additionally, the conductivity of the lake also increases at 5 meters below surface levels, remaining constant at 13 uS/cm, increasing up to 60 uS/cm at the bottom of the lake. Spectral characteristics measured appeared to be relatively constant via colour absorbance values, however, chlorophyll-a concentrations increased with depth. Oxygen concentration and saturation percentage values had also appeared to drop drastically at 4 meters, where the values would drop to near 0. The pH values measured at the lake showed a slight decrease in pH, where the surface pH was 7.45 and dropped to 5.5 near the bottom of the lake. Alkalinity values yielded from the lake increased with depth, where surface

level to the 3 meter sample showed a concentration of 8.5 mg/L, and increased towards 35 mg/L at the bottom of the lake. Total phosphorus, total iron, ferrous iron and sulfide concentrations represented a gradual increase in concentration with lake depth, while sulfate and nitrate concentrations remained relatively constant.

4. Discussion

4.1 Interpretation

4.1.1 Lac Cromwell

Lac Cromwell, with a depth of 9 meters, is a dystrophic lake, with high amounts of dissolved organic carbons with humic acids. Lac Cromwell was stratified based off of the temperature and oxygen profiles, where a notable dip is indicated in the graphs at the metalimnion-hypolimnion interface. Based off of figure 1, Lac Cromwell represents a eutrophic lake, with a clinograde oxygen profile. This is characterized by the depletion of O₂ saturation in the hypolimnion zone. This can suggest that oxygen depletion was dominated by the microbial decomposition of organic matter at the bed of the lake (Marcé et al., 2024).

The electrical conductivity within the vertical profile Lac Cromwell was represented by micro-Siemens per cm. This represents the dissolved ions within the water of the lake, and with the Lac Cromwell, the increase in conductivity within the hypolimnion was not typically representative of the temperature as well. Conductivity typically increases with temperature, however in this lake (Pawlowicz & Yerubandi, 2024), the conductivity and temperature seem to have an inverse relationship where as the temperature decreased, conductivity increased. This

may indicate the presence of an abundance of dissolved organic matter within the hypolimnion zone of Lac Cromwell.

Total phosphorus levels in Lac Cromwell appeared to increase in depth. This is represented by the high amounts of dissolved organic matter and the sediments at the bottom of the lake, which further supports the characterization of the lake (Dillon & Molot, 2024). When observing ferrous and total iron an increase within the lake depth. This may be explained by the abundance of humic acid within the lake content, which reduces ferric iron into ferrous iron via microbial activity. Total iron can also be attributed towards suspended iron within the water column, as well as iron within the lake sediments (Kritzberg, 2024).

4.1.2 Lac Croche

Lac Croche is characterized as an Oligotrophic soft water lake with a small drainage basin, receiving water from rainfall. Lac Croche is the deepest lake, with a depth of approximately 12 m. Similarly to Lac Cromwell, Lac Croche presented notable dips in oxygen concentration and a clinograde profile. Additionally, the temperature gradient and conductivity were similar as well, where conductivity had increased with depth. This trend in conductivity shows less of an increase in concentration in comparison, however. This can be explained by a lower dissolved organic matter abundance, where dissolved ions were present among the lake. The lake profile does however appear to be stratified based off of the oxygen saturation percentage and concentration, where the hypolimnion zone begins around 6 meters under the surface of water.

The anomaly present in the ferrous iron panel of the multi-figure (refer to figure 2.) can be explained by the presence of ions from the sediment-water interface. Ferrous iron can also be reduced by the precipitation of insoluble iron-sulfides in the hypolimnion (Kritzberg, 2024). This

anomaly can further be explained by the trend in the total iron panel, where concentrations gradually increase with depth as well.

A major source of phosphorus derives from organic matter and adsorption through release from sediments. Lac Croche appears to possess higher productivity among the lake, which coincides with the considerably high concentration of total phosphorus among the hypolimnion region of the lake (Dillon & Molot, 2024).

Nitrate concentrations remained consistently among the depth profile of the lake. This can suggest that the enrichment from nitrification and denitrification processes worked in tandem, which appears to characterize the lake in a more mesotrophic setting rather than an oligotrophic lake.

4.1.3 Lac Triton

Lac Triton is a Mesotrophic Lake with no apparent flow. This lake is the shallowest lake with a depth of 4 meters. The results based off of figure 3. Represent relatively static thermal, chemical, and spectral characteristics. The data generally does not represent any stratification among the lake, however the state cannot determine whether or not this lake mixes or not due to the observation of the lake only at one point during the year.

Based off of the temperature profile of Lac Triton, the temperature remained the same for the depth of the lake. Similarly with conductivity, and O₂ concentrations and percentage saturations, these values all remained consistent throughout the water column. The consistency in conductivity represents the same amounts of dissolved organic matter throughout the lake. Additionally, there is no depletion detected among the profile, which was neither clinograde nor orthograde. This can suggest that there is lower productivity among Lac Triton.

A noticeable feature of the chemical characteristics within the water column appear to be based off of the sulfide, total iron, and ferrous iron concentrations which all contribute to relative productivity.

4.1.3 Lac Geai

Lac Geai has been previously characterized as a small, dystrophic, bog-like lake with an abundance of dissolved organic matter. Lac Geai has a depth of 9 meters below surface level. Mentioned previously, the conductivity of the lake had increased with depth, while temperature and O₂ levels depleted. This is conducive to a highly stratified, clinograde profile lake. Lac Geai presents high productivity in the lake.

pH values indicated in figure 4 were particularly intriguing since Lac Geai is considered as a bog-like lake. Bogs characteristically have lower pH values, however, surface pH values recorded were relatively high, with a value of 7.45. This is unusual for a bog-like lake and could be explained by the presence of carbonate rocks surrounding the area and precipitating into the surficial water in the lake. This corresponds to the alkalinity, where the surface alkalinity values are considerably lower than the hypolimnion values.

Total phosphorus, ferrous iron, total iron, and sulfide concentrations all increased in depth. Notably, a strong sulfur smell was observed in the field after collecting water samples lower in depth. This indicates the presence of an abundance of hydrogen sulfide. Phosphorus and nitrate levels were considerably high in the hypolimnion regions suggesting an abundance of dissolved organic matter among the lake (Burford & Lu, 2024; Dillon & Molot, 2024). These previously mentioned characteristics were reminiscent of a bog-like, dystrophic lake.

4.2 Limitations of the Study

A limitation of the study that should be mentioned is that this study is based off of one visit to the study. During this visit, all samples were collected in one day. This limits the identification of any mixing / stratification statuses with lakes such as Lac Triton. Additionally, this represents just a snap shot in time for the current season that the data was collected and analyzed at (late summer/early fall).

An additional limitation to the study is the collection of ORP (oxidation-reduction potential) data. This data would give valuable insight into the status of a lakes microbial activity, and interactions with nutrients and chemicals such as irons and sulfides. It would also help further support oxic/anoxic conditions.

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Appendix

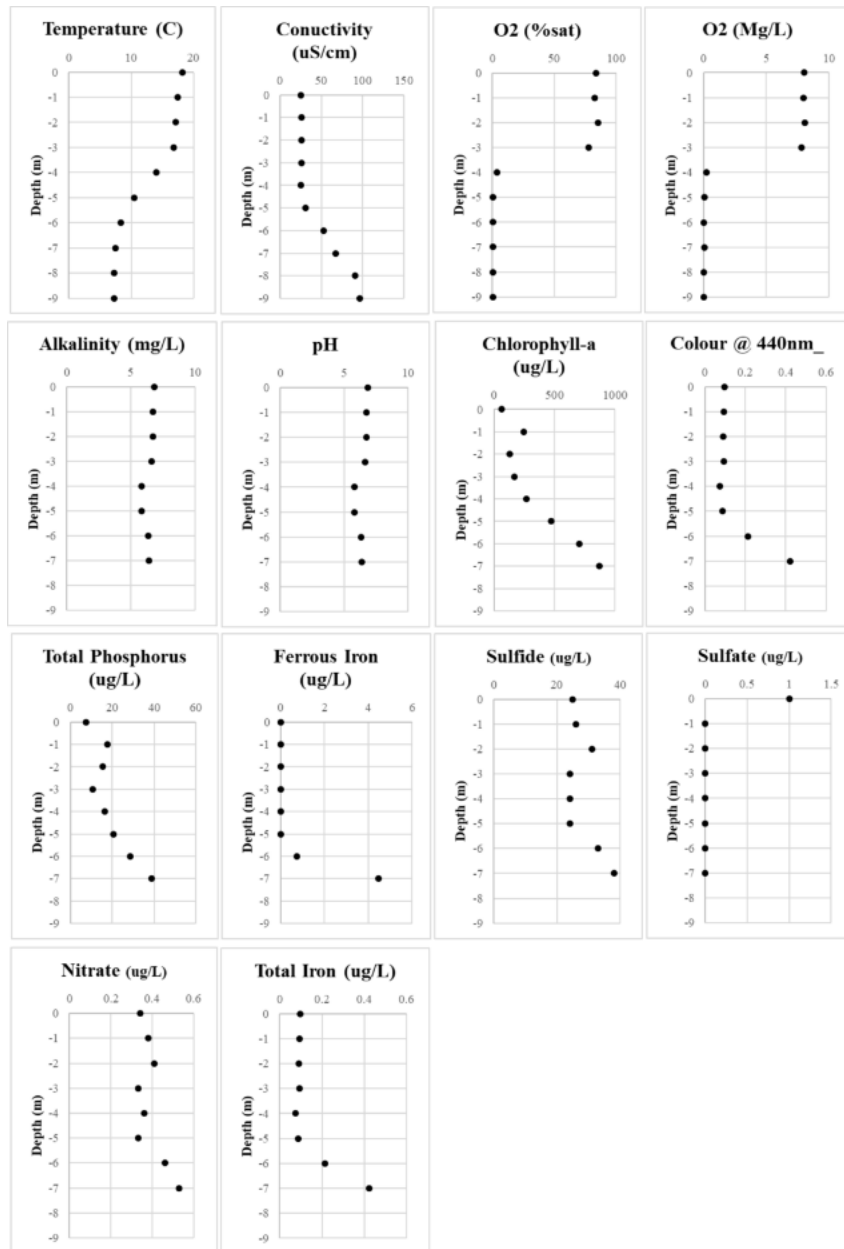


Figure 1. Multi-figure of Lac Cromwell. Chemical, spectral, and thermal characteristics of Lac Cromwell. Measurements include temperature, conductivity, O₂ percent saturation and concentration, alkalinity, pH, chlorophyll-a, colour, total phosphorus, ferrous iron, sulfide, sulfide, sulfate, nitrate, and total iron.

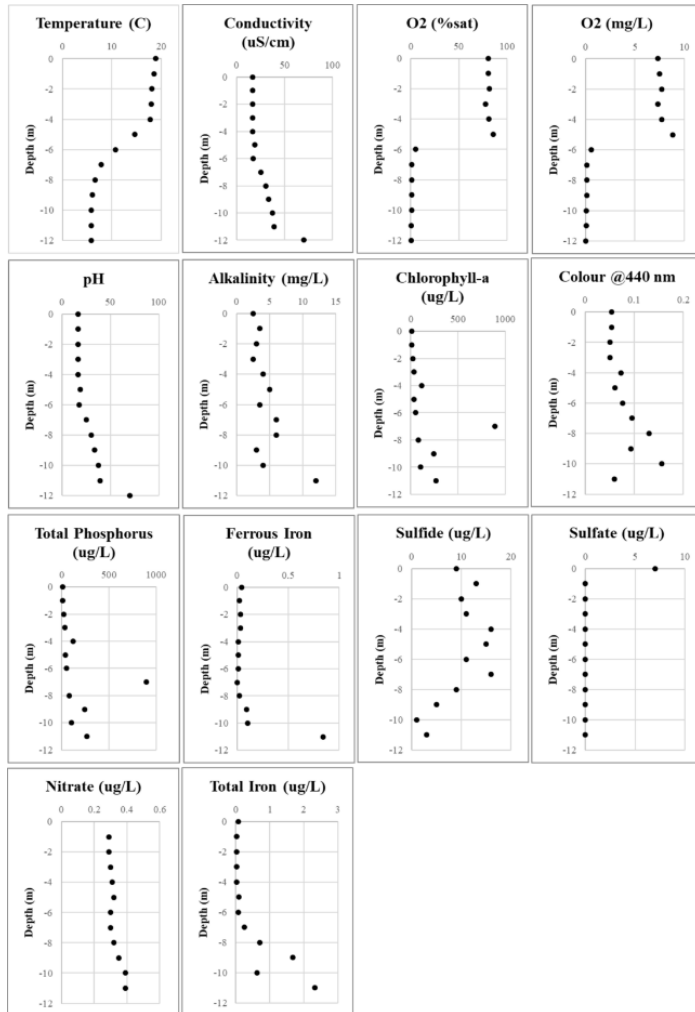


Figure 2. Multi-figure of Lac Croche. Chemical, spectral, and thermal characteristics of Lac Croche. Measurements include temperature, conductivity, O₂ percent saturation and concentration, alkalinity, pH, chlorophyll-a, colour, total phosphorus, ferrous iron, sulfide, sulfate, nitrate, and total iron.

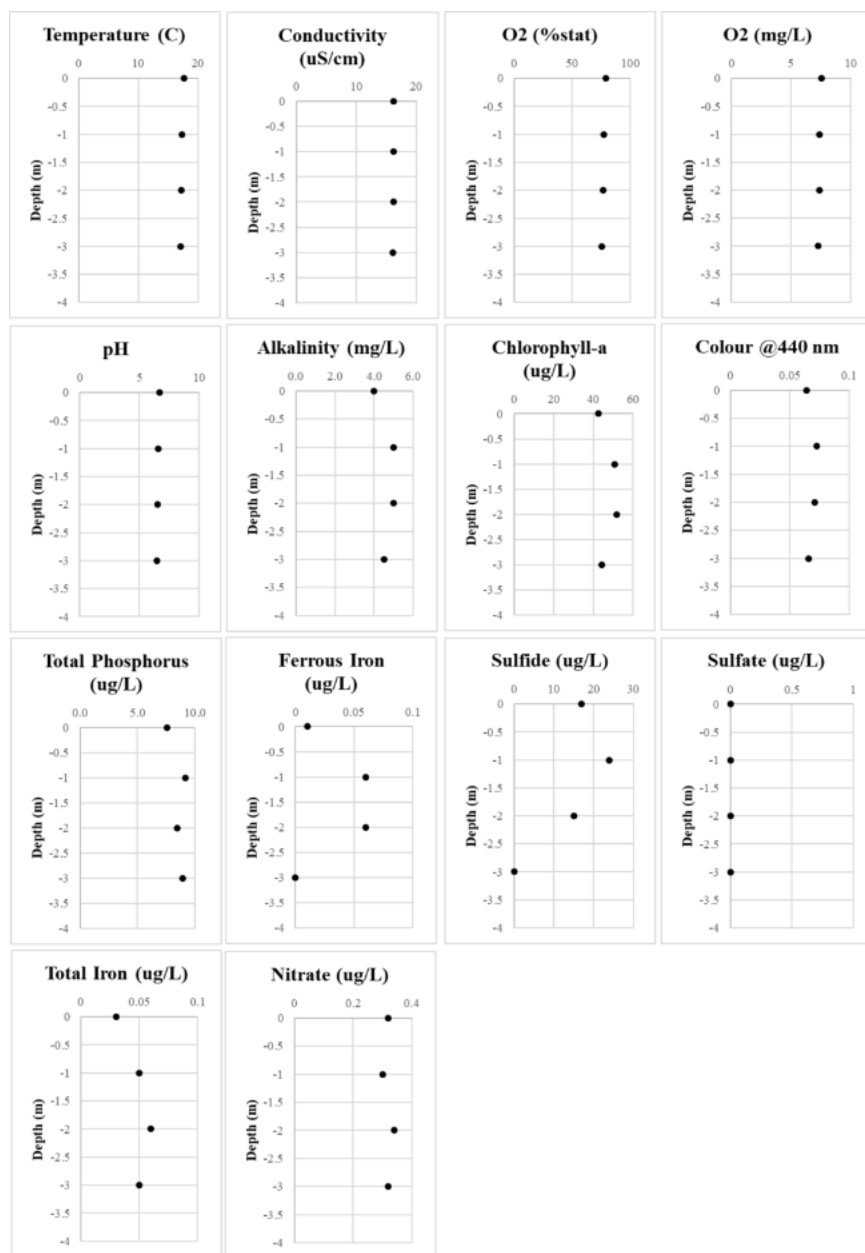


Figure 3. Multi-figure of Lac Triton. Chemical, spectral, and thermal characteristics of Lac Triton. Measurements include temperature, conductivity, O₂ percent saturation and concentration, alkalinity, pH, chlorophyll-a, colour, total phosphorus, ferrous iron, sulfide, sulfate, nitrate, and total iron.

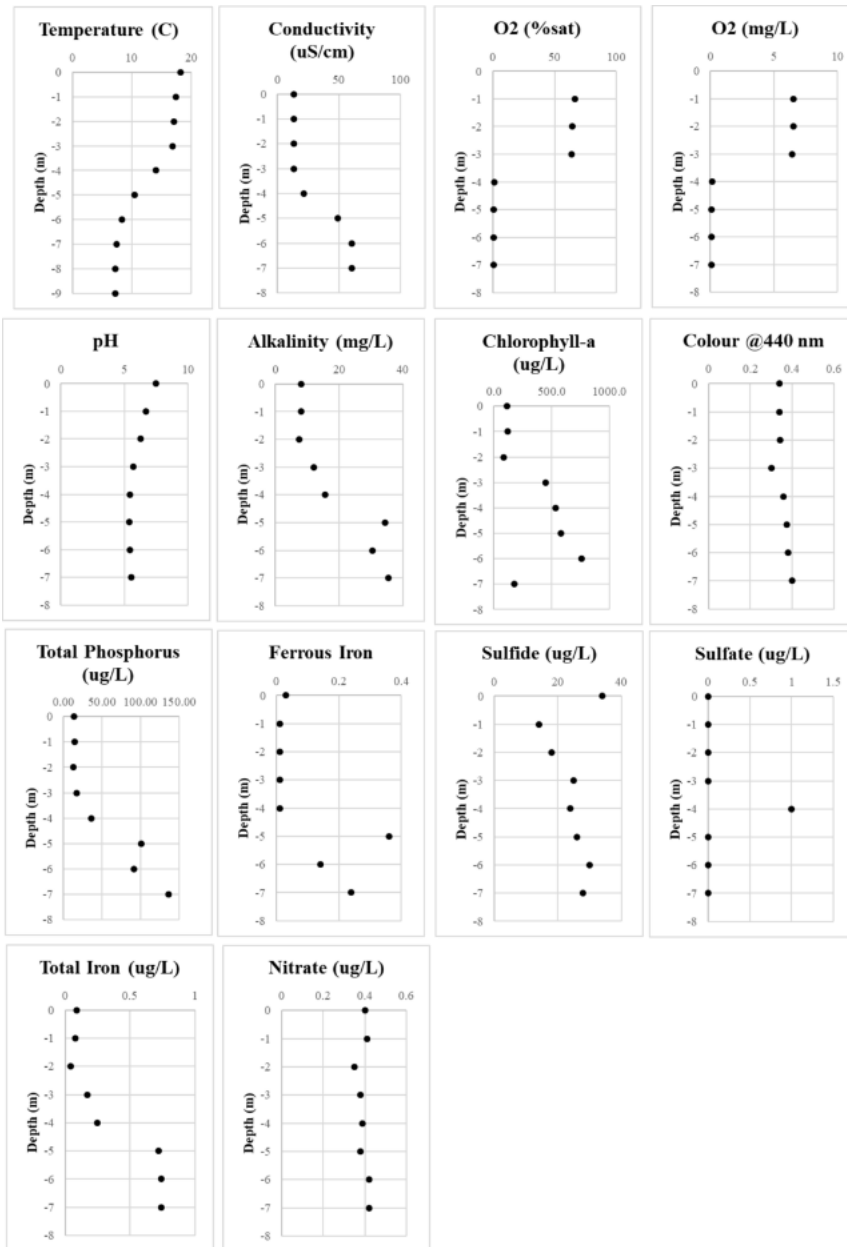


Figure 4. Multi-figure of Lac Geai. Chemical, spectral, and thermal characteristics of Lac Geai. Measurements include temperature, conductivity, O₂ percent saturation and concentration, alkalinity, pH, chlorophyll-a, colour, total phosphorus, ferrous iron, sulfide, sulfate, nitrate, and total iron.